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PROJECTION-DRIVEN SHARP MOMENT THRESHOLDS FOR PAIRWISE U-STATISTICS: ASYMPTOTIC NORMALITY AND BOOTSTRAP FAILURE UNDER HEAVY TAILS

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We develop a pairwise U-statistic framework for variance and kurtosis inference under heavy-tailed sampling. The sample pairwise difference variance SPDV_n is a non-degenerate second-order U-statistic whose asymptotic variance is governed by the fourth central moment. We further introduce the quartic pairwise dispersion functional θ_4 with U-statistic estimator SPFD_n, satisfying the explicit identity $\mu_4 = 6\theta_4 - 3\sigma^4$. Hoeffding projections yield sharp CLT thresholds: fourth and eighth moments for $\sqrt{n}$-normality of SPDV_n and SPFD_n. These thresholds refine general U-statistic bootstrap results by translating kernel degree into explicit minimal moment conditions for variance asymptotics and resampling validity. For the studentized bootstrap of SPFD_n, twelfth moments are necessary and sufficient within regularly varying families with tail index $\alpha > 8$, establishing a formal uniform impossibility result over the range $\alpha \in (8, 12)$. Failure occurs through projection tail divergence, where a single extreme observation dominates the resampling distribution. Practically, our results (i) delineate when normal approximations remain reliable, (ii) identify the regimes where percentile versus studentized bootstrap should be preferred, and (iii) justify pairwise, mean-centering-free estimators as robust alternatives to classical mean-centered moment estimators in heavy-tailed settings.

More broadly, we establish these asymptotic and bootstrap thresholds sharply for the quadratic ($k = 1$) and quartic ($k = 2$) pairwise kernels, prove the corresponding projection-growth principle for the even pairwise family at the level of asymptotic normality, and formulate the associated $6k$ studentized-bootstrap scaling law as an open conjecture supported by the quadratic and quartic cases.

1. Introduction. Higher-order moments play a central role in variance estimation, Gaussian approximation, and resampling-based inference, but their behavior becomes fragile under heavy-tailed sampling. In particular, fourth-moment methods are highly sensitive to extreme observations, and this sensitivity is often obscured when variance and kurtosis are expressed solely through mean-centered empirical moments. Although the sample variance admits a classical U-statistic representation, the fourth-order quantities governing its asymptotic variability and bootstrap calibration are usually not analyzed through the projection structure of that representation. As a result, the mechanism by which heavy tails degrade normal approximation and bootstrap performance remains only partially explicit.

A pairwise formulation isolates this mechanism more transparently. Because pairwise kernels measure dispersion through differences between observations rather than repeated centering by the sample mean, they expose the first-order Hájek projection directly and make the

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relevant moment requirements interpretable in terms of the underlying distribution. This perspective is especially useful under heavy tails, where classical empirical kurtosis is unstable and its variability propagates into variance estimation, studentization, and coverage accuracy. In particular, the pairwise formulation separates the fluctuation of the first-order projection from the instability introduced by repeated mean centering, thereby making the dominant term in the Hoeffding decomposition more transparent.

Our analysis builds on the classical theory of non-degenerate U-statistics and their Hoeffding decompositions [8, 11, 9], together with bootstrap theory for U- and V-statistics [2, 1, 4, 7]. That theory gives general conditions for asymptotic normality and bootstrap validity, typically formulated in terms of integrability assumptions on the kernel or its projections. For the polynomial pairwise kernels studied here, these abstract conditions can be translated into explicit moment requirements on the underlying distribution. This translation yields kernel-specific sharp thresholds that are not transparent in the general theory, and makes it possible to identify the precise points at which Gaussian approximation and bootstrap validity break down under heavy-tailed sampling. In particular, these thresholds are determined by the growth of the first-order projection, providing a unified structural explanation across asymptotic normality and resampling regimes.

For the sample variance, classical theory already implies that finite fourth moments are necessary and sufficient for $\sqrt{n}$-asymptotic normality. While these results are implicit in the general theory of U-statistics, their interpretation through a unified projection-based mechanism that extends to higher-order pairwise functionals and bootstrap procedures has not been systematically developed. In particular, for quadratic and quartic pairwise kernels, the correspondence between kernel degree, projection growth, and bootstrap breakdown has not been isolated in a sharp form. The present paper identifies that correspondence for the variance and quartic-dispersion settings studied here, and proves necessity of the resulting thresholds within a broad regularly varying heavy-tail class.

Relation to classical theory. The underlying Hoeffding decomposition and the general bootstrap theory for non-degenerate U-statistics are classical [8, 11, 1, 4]. The contribution of the present paper is to translate these abstract conditions into explicit moment thresholds for the pairwise polynomial statistics studied here, and to show in the quartic case that the studentized bootstrap exhibits a sharp heavy-tail breakdown boundary within a regularly varying class. Thus, the novelty lies not in the general U-statistic machinery itself, but in the explicit identification of sharp moment thresholds, the sharp impossibility result for studentized bootstrap validity within a regularly varying class, and the projection-growth mechanism linking these phenomena.

We work in the nonparametric i.i.d. model and study two pairwise statistics. The first is the sample pairwise difference variance

$$\mathrm{SPDV}_n = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \frac{(X_i - X_j)^2}{2},$$

which is algebraically identical to the usual sample variance and whose asymptotic behavior is governed by the fourth moment through its first-order projection. The second is the quartic pairwise functional

$$\theta_4 = \mathbb{E}\left[\frac{(X_1 - X_2)^4}{12}\right], \qquad \mathrm{SPFD}_n = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \frac{(X_i - X_j)^4}{12},$$

which defines a non-degenerate second-order U-statistic linked to the fourth central moment through the exact identity

$$\mu_4 = 6\theta_4 - 3\sigma^4.$$

Our aim is to determine the sharp moment conditions under which SPDV_n and SPFD_n are $\sqrt{n}$-asymptotically normal, and under which their nonparametric bootstrap analogues remain valid, both in percentile and studentized form.

The main contributions are as follows. First, we provide an explicit projection-level analysis that translates the abstract conditions of general non-degenerate U-statistic theory into explicit, kernel-specific moment thresholds. For the quadratic and quartic pairwise kernels, this recovers fourth- and eighth-moment thresholds for $\sqrt{n}$-asymptotic normality, and for the even pairwise family it proves the corresponding $4k$ projection-growth principle at the level of asymptotic normality.

Second, for the quartic statistic SPFD_n, we establish a sharp threshold for studentized bootstrap validity: finite twelfth moments are sufficient for consistency and are also necessary within a broad class of regularly varying heavy-tailed distributions. This yields a sharp characterization of bootstrap breakdown driven by extreme observations within that class.

Third, we identify projection dominance as the structural mechanism governing both asymptotic normality and bootstrap validity: when the first-order projection dominates, Gaussian and bootstrap approximations hold, whereas failure of this dominance leads to non-Gaussian behavior and resampling instability.

Finally, the exact identity $\mu_4 = 6\theta_4 - 3\sigma^4$ provides a bridge between pairwise quartic dispersion and classical fourth-moment inference, yielding mean-centering-free estimators and pairwise variance formulas that remain interpretable under heavy-tailed sampling.

The remainder of the paper is organized as follows. Section 2 introduces the pairwise difference framework and re-expresses the sample variance as a second-order U-statistic, providing the projection baseline for the analysis that follows. Section 3 then develops the quartic pairwise dispersion functional, derives its exact Hoeffding projection structure, and links it to classical fourth-moment inference through an explicit pairwise identity. Section 4 establishes the sharp moment thresholds for asymptotic normality, and Section 5 investigates bootstrap validity, with particular emphasis on the distinction between percentile and studentized procedures and the sharp twelfth-moment threshold governing the quartic case. Section 6 presents a numerical illustration of the resulting threshold phenomena under heavy-tailed sampling, and Section 7 closes with implications, extensions, and open questions. The proofs of the main results are deferred to Appendix A. Additional simulation details, extended numerical tables, and supplementary figures are reported in the supplementary material. Reproducible R code for the numerical experiments is available at https://github.com/asrivas18/AoS-ssbvi.

2. Pairwise Difference Framework.

2.1. *Notation and setup.* Throughout, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} F$, where F is a distribution on $\mathbb{R}$ with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \mathrm{Var}(X) > 0$. For $k \geq 2$, we write

$$\mu_k = \mathbb{E}[(X-\mu)^k]$$

for the kth central moment, and we use absolute moment conditions such as $\mathbb{E}[|X|^k] < \infty$ and $\mathbb{E}[|X-\mu|^k] < \infty$ interchangeably, since finiteness of one is equivalent to finiteness of the other. All asymptotic statements are taken as $n \to \infty$, and convergence in distribution and probability are denoted by $\xrightarrow{d}$ and $\xrightarrow{p}$, respectively.

The classical sample variance admits an exact pairwise representation, which makes it natural to study variance estimation through the theory of non-degenerate second-order U-statistics. This formulation isolates the first-order Hájek projection explicitly and thereby makes the relevant moment requirement transparent.

2.2. *The pairwise variance statistic.*

DEFINITION 1 (Sample pairwise difference variance). *Define*

$$\mathrm{SPDV}_n = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \frac{(X_i - X_j)^2}{2}. \tag{1}$$

The statistic SPDV_n is simply the sample variance written in pairwise form. Although algebraically equivalent to the usual mean-centered expression, the pairwise formulation makes the projection structure explicit and serves as the baseline for the higher-order analysis developed later.

PROPOSITION 1 (Pairwise representation of variance). *For any real numbers $x_1, \dots, x_n$,*

$$\frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} \frac{(x_i - x_j)^2}{2} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2, \qquad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Consequently,

$$\mathrm{SPDV}_n = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

PROOF. The identity follows from the classical algebraic relation

$$\sum_{1\le i<j\le n} (x_i - x_j)^2 = n \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Dividing both sides by $2\binom{n}{2} = n(n-1)$ yields the result. □

REMARK 1 (Comparison with mean-centered representations). *Although Proposition 1 shows that SPDV_n is algebraically identical to the sample variance, its pairwise representation is asymptotically more revealing than the usual mean-centered form. In particular, the leading stochastic term is expressed directly through the projection $\psi_2(X)$, so the fourth-moment threshold arises transparently from the growth of the projection itself. This perspective is especially useful under heavy-tailed sampling, where the interaction between extreme observations and empirical centering can obscure the mechanism governing fluctuation.*

Thus variance may be viewed as the average squared distance between two observations. This interpretation places the sample variance directly within the standard framework of non-degenerate second-order U-statistics.

THEOREM 1 (SPDV as a second-order U-statistic). *Let*

$$h_2(x, y) = \frac{(x-y)^2}{2}.$$

Then

$$\mathrm{SPDV}_n = \frac{1}{\binom{n}{2}} \sum_{1\le i<j\le n} h_2(X_i, X_j)$$

is a second-order U-statistic with parameter

$$\theta_2 = \mathbb{E}[h_2(X_1, X_2)] = \sigma^2.$$

Moreover, if $\sigma^2 > 0$, then the kernel is non-degenerate.

PROOF. By independence and symmetry,

$$\mathbb{E}[h_2(X_1, X_2)] = \frac{1}{2}\mathbb{E}[(X_1 - X_2)^2] = \frac{1}{2}\operatorname{Var}(X_1 - X_2) = \frac{1}{2}(2\sigma^2) = \sigma^2.$$

The first-order Hájek projection is

$$\psi_2(x) = \mathbb{E}[h_2(x, X_2)] - \sigma^2.$$

Using $\mathbb{E}[(x - X_2)^2] = (x - \mu)^2 + \sigma^2$, we obtain

$$\psi_2(x) = \frac{(x-\mu)^2 - \sigma^2}{2}. \tag{2}$$

If $\sigma^2 > 0$, then $(X - \mu)^2$ is non-constant almost surely, so $\operatorname{Var}(\psi_2(X)) > 0$, and the kernel is non-degenerate. □

By standard U-statistic theory, Theorem 1 yields the Hoeffding decomposition

$$\mathrm{SPDV}_n - \sigma^2 = \frac{2}{n}\sum_{i=1}^{n} \psi_2(X_i) + R_n, \qquad R_n = O_p(n^{-1}), \tag{3}$$

where ψ_2 is given by (2). In particular,

$$\sqrt{n}\, R_n = o_p(1)$$

for this non-degenerate second-order kernel [8, 11, 9].

The decomposition (3) is the key structural fact for the rest of the paper: the asymptotic behavior of SPDV_n is determined by the projection $\psi_2(X)$, whose variance exists exactly when the fourth moment of X is finite.

2.3. *Asymptotic normality and sharpness.*

THEOREM 2 (CLT for SPDV_n). *If $\mathbb{E}[|X - \mu|^4] < \infty$, then*

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) \xrightarrow{d} \mathcal{N}\big(0, \mu_4 - \sigma^4\big).$$

PROOF. From (3),

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i) + \sqrt{n}\, R_n.$$

Since

$$\psi_2(X) = \frac{(X-\mu)^2 - \sigma^2}{2},$$

we have

$$\operatorname{Var}(\psi_2(X)) = \frac{1}{4}\operatorname{Var}\big((X-\mu)^2\big) = \frac{1}{4}\left(\mu_4 - \sigma^4\right).$$

Under $\mathbb{E}[|X - \mu|^4] < \infty$, the variance of $\psi_2(X)$ is finite, so the classical central limit theorem gives

$$\frac{2}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i) \xrightarrow{d} \mathcal{N}(0, 4\operatorname{Var}(\psi_2(X))) = \mathcal{N}\big(0, \mu_4 - \sigma^4\big).$$

Since $\sqrt{n}\, R_n = o_p(1)$, Slutsky's theorem completes the proof. □

Thus the asymptotic variance of the sample variance is governed by the fourth central moment through the projection variance. In particular, $\mathbb{E}[|X-\mu|^4] < \infty$ characterizes the regime in which the classical $\sqrt{n}$-Gaussian approximation applies; outside this regime, alternative non-Gaussian limits or divergence may occur.

PROPOSITION 2 (Necessity of the fourth moment). *If $\mathbb{E}|X-\mu|^4 = \infty$, then the sequence*

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2)$$

cannot converge in distribution to a non-degenerate Gaussian law.

PROOF. By the Hoeffding decomposition,

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) = \frac{2}{\sqrt{n}} \sum_{i=1}^{n} \psi_2(X_i) + o_p(1).$$

If $\mathbb{E}|X-\mu|^4 = \infty$, then $\mathrm{Var}(\psi_2(X)) = \infty$, since

$$\psi_2(x) = \frac{(x-\mu)^2 - \sigma^2}{2}$$

is a quadratic polynomial in $(x-\mu)$ with nonzero leading coefficient. Therefore the normalized sum

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_2(X_i)$$

cannot converge in distribution to a non-degenerate Gaussian law. Since the remainder term is $o_p(1)$ at the same scale, the same conclusion holds for

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2).$$

□

Proposition 2 identifies the fourth moment as the sharp threshold for classical $\sqrt{n}$-normality of SPDV_n. This quadratic case serves as the baseline for the quartic pairwise dispersion functional developed next, where the same projection-based logic yields a sharper and more demanding eighth-moment threshold.

3. Pairwise Fourth-Order Dispersion.

3.1. *Definition and identity.* The pairwise variance statistic SPDV_n developed in Section 2 captures dispersion through squared pairwise differences. To analyze the higher-order structure governing its variability, and in particular the behavior of associated bootstrap procedures, we introduce a quartic analogue that isolates fourth-order effects at the kernel level.

DEFINITION 2 (Pairwise fourth-order dispersion functional). *Define*

$$\theta_4 = \mathbb{E}\left[\frac{(X_1 - X_2)^4}{12}\right],$$

where $X_1, X_2 \overset{\text{iid}}{\sim} F$, and let

$$\mathrm{SPFD}_n = \frac{1}{\binom{n}{2}} \sum_{1 \le i < j \le n} \frac{(X_i - X_j)^4}{12}. \tag{4}$$

Thus SPFD_n is the natural quartic analogue of SPDV_n, obtained by replacing squared pairwise differences with fourth powers. The normalization by 12 is chosen so that θ_4 is linked to the fourth central moment by a simple exact identity.

THEOREM 3 (Pairwise representation of the fourth central moment). *Assume* $\mathbb{E}[|X-\mu|^4] < \infty$. *Then*

$$\theta_4 = \frac{\mu_4}{6} + \frac{\sigma^4}{2}, \tag{5}$$

and hence

$$\mu_4 = 6\theta_4 - 3\sigma^4. \tag{6}$$

Consequently,

$$\widehat{\mu}_4^{\mathrm{pair}} := 6\,\mathrm{SPFD}_n - 3\,(\mathrm{SPDV}_n)^2 \tag{7}$$

provides a fully pairwise estimator of the fourth central moment.

PROOF. Let $Z_i = X_i - \mu$, so that Z_1, Z_2 are independent with mean zero and variance σ^2. Expanding,

$$(X_1 - X_2)^4 = (Z_1 - Z_2)^4 = Z_1^4 + Z_2^4 + 6Z_1^2 Z_2^2 - 4Z_1^3 Z_2 - 4Z_1 Z_2^3.$$

Taking expectations and using independence together with $\mathbb{E}[Z_i] = 0$ gives

$$\mathbb{E}[(X_1 - X_2)^4] = 2\mu_4 + 6\sigma^4.$$

Dividing by 12 yields

$$\theta_4 = \frac{2\mu_4 + 6\sigma^4}{12} = \frac{\mu_4}{6} + \frac{\sigma^4}{2},$$

which proves (5). Rearranging gives (6), and substituting SPFD_n and SPDV_n for their population counterparts yields (7). □

The identity (6) closes the inferential loop between pairwise variance and quartic dispersion. In particular, it shows that fourth-moment estimation can be expressed without explicit sample-mean centering, which is especially useful under heavy-tailed sampling.

REMARK 2 (Relation to classical fourth-moment estimation). *Classical estimators of* μ_4 *are based on empirical powers of* $(X_i - \bar{X})$, *whose behavior can be unstable under heavy-tailed sampling because the sample mean is itself sensitive to extreme observations. In contrast, representation* (6) *expresses the fourth central moment entirely in terms of pairwise* U*-statistics, separating the quartic dispersion component from the variance correction term. This formulation makes the projection structure explicit and provides a more stable basis for higher-order inference under heavy-tailed distributions.*

3.2. *Projection structure.* We next place SPFD_n within the standard second-order U-statistic framework.

THEOREM 4 (SPFD_n as a second-order U-statistic). *Assume* $\mathbb{E}[|X-\mu|^4] < \infty$, *and define*

$$h_4(x, y) = \frac{(x-y)^4}{12}.$$

Then

$$\mathrm{SPFD}_n = \frac{1}{\binom{n}{2}} \sum_{1 \le i < j \le n} h_4(X_i, X_j)$$

is a second-order U-statistic with parameter

$$\mathbb{E}[h_4(X_1, X_2)] = \theta_4.$$

Moreover, if $\sigma^2 > 0$, then $\psi_4(X)$ is non-constant and has strictly positive variance, so the kernel is non-degenerate in the sense of Hoeffding.

PROOF. The representation follows directly from Definition 2, and the parameter identity is exactly Theorem 3. The expression for the first-order projection is derived in Lemma 1. Since its leading term is quartic in $(x - \mu)$, the projection is non-constant whenever $\sigma^2 > 0$, and therefore has strictly positive variance. Hence the kernel is non-degenerate in the sense of Hoeffding. □

LEMMA 1 (Exact Hájek projection for SPFD_n). *Let $\kappa_3 = \mathbb{E}[(X - \mu)^3]$. Then the first-order projection of the kernel h_4 is*

$$\psi_4(x) = \frac{(x-\mu)^4}{12} + \frac{(x-\mu)^2\sigma^2}{2} - \frac{(x-\mu)\kappa_3}{3} - \frac{\mu_4}{12} - \frac{\sigma^4}{2}. \tag{8}$$

PROOF. Let $Y \sim F$ be independent of x. Writing $W = Y - \mu$, we have

$$x - Y = (x - \mu) - W.$$

Therefore,

$$\mathbb{E}[(x - Y)^4] = (x-\mu)^4 + 6(x-\mu)^2\mathbb{E}[W^2] - 4(x-\mu)\mathbb{E}[W^3] + \mathbb{E}[W^4],$$

since $\mathbb{E}[W] = 0$. Using $\mathbb{E}[W^2] = \sigma^2$, $\mathbb{E}[W^3] = \kappa_3$, and $\mathbb{E}[W^4] = \mu_4$, we obtain

$$\mathbb{E}[h_4(x, Y)] = \frac{(x-\mu)^4 + 6(x-\mu)^2\sigma^2 - 4(x-\mu)\kappa_3 + \mu_4}{12}.$$

Subtracting

$$\theta_4 = \frac{\mu_4}{6} + \frac{\sigma^4}{2}$$

from both sides yields (8). □

The representation (8) shows that ψ_4 is a polynomial of degree four in $(x - \mu)$, with leading term $(x - \mu)^4/12$. Thus the tail behavior of the projection is governed by the fourth power of the underlying variable, and this determines the moment conditions required for asymptotic normality and, later, for bootstrap validity.

THEOREM 5 (Hoeffding decomposition for SPFD_n). *Assume $\mathbb{E}[|X - \mu|^4] < \infty$. Then*

$$\mathrm{SPFD}_n - \theta_4 = \frac{2}{n}\sum_{i=1}^{n} \psi_4(X_i) + R_{4,n}, \qquad R_{4,n} = O_p(n^{-1}), \tag{9}$$

where ψ_4 is given by (8). In particular,

$$\sqrt{n}\, R_{4,n} = o_p(1).$$

PROOF. This is the standard Hoeffding decomposition for a non-degenerate second-order U-statistic with kernel h_4; see [8, 11, 9]. □

The decomposition (9) shows that the asymptotic behavior of SPFD_n is determined by the projection $\psi_4(X)$. In contrast to the quadratic case, the quartic leading term makes the variance of the linear component sensitive to eighth moments.

The asymptotic variance of the linear term in (9) is

$$\tau_4^2 = 4\,\mathrm{Var}(\psi_4(X)). \tag{10}$$

A full expansion of τ_4^2 in central moments up to order eight is given in the appendix.

REMARK 3 (Symmetric simplification). *If the distribution of X is symmetric about μ, then $\kappa_3 = 0$, and the projection simplifies to*

$$\psi_4(x) = \frac{(x-\mu)^4}{12} + \frac{(x-\mu)^2\sigma^2}{2} - \frac{\mu_4}{12} - \frac{\sigma^4}{2}. \tag{11}$$

Accordingly, the variance expression for τ_4^2 simplifies as well; the explicit symmetric formula is recorded in the appendix.

3.3. *Sharp eighth-moment threshold.* The decomposition (9) reduces the asymptotic analysis of SPFD_n to the integrability properties of the projection $\psi_4(X)$.

REMARK 4 (Moment equivalence). *Since $\psi_4(x)$ is a polynomial in $(x-\mu)$ with leading term $(x-\mu)^4/12$, and all lower-order coefficients are finite under $\mathbb{E}[|X-\mu|^4] < \infty$, its square is a polynomial with leading term $(x-\mu)^8/144$. It follows that $\mathrm{Var}(\psi_4(X)) < \infty$ if and only if $\mathbb{E}[|X-\mu|^8] < \infty$.*

THEOREM 6 (CLT for SPFD_n). *If $\mathrm{Var}(\psi_4(X)) < \infty$, equivalently if $\mathbb{E}[|X-\mu|^8] < \infty$, then*

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4) \xrightarrow{d} \mathcal{N}(0, \tau_4^2),$$

where τ_4^2 is defined by (10).

PROOF. By Theorem 5,

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4) = \frac{2}{\sqrt{n}} \sum_{i=1}^{n} \psi_4(X_i) + \sqrt{n}\,R_{4,n}.$$

Under $\mathrm{Var}(\psi_4(X)) < \infty$, the classical central limit theorem yields

$$\frac{2}{\sqrt{n}} \sum_{i=1}^{n} \psi_4(X_i) \xrightarrow{d} \mathcal{N}(0, 4\,\mathrm{Var}(\psi_4(X))) = \mathcal{N}(0, \tau_4^2).$$

Since $\sqrt{n}\,R_{4,n} = o_p(1)$, Slutsky's theorem completes the proof. □

PROPOSITION 3 (Necessity of the eighth moment). *If $\mathbb{E}|X-\mu|^8 = \infty$, then the sequence*

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4)$$

cannot converge in distribution to a non-degenerate Gaussian law.

PROOF. By the Hoeffding decomposition,

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4) = \frac{2}{\sqrt{n}} \sum_{i=1}^{n} \psi_4(X_i) + o_p(1).$$

If $\mathbb{E}|X-\mu|^8 = \infty$, then $\mathrm{Var}(\psi_4(X)) = \infty$ by Remark 4, since $\psi_4(x)$ is a polynomial in $(x-\mu)$ with quartic leading term. Therefore the normalized sum

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_4(X_i)$$

cannot converge in distribution to a non-degenerate Gaussian law. Since the remainder term is $o_p(1)$ at the same scale, the same conclusion holds for

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4).$$

□

Theorem 6 together with Proposition 3 shows that the eighth moment condition is the sharp threshold for $\sqrt{n}$-asymptotic normality of SPFD_n. Compared with the quadratic case, where fourth moments suffice, this doubling reflects the quartic growth of the first-order projection and illustrates the direct correspondence between kernel degree and moment requirements.

4. Moment Thresholds and Variance Links. This section consolidates two consequences of the pairwise framework developed in Sections 2 and 3. First, the quartic pairwise statistic yields a direct estimator of the asymptotic variance of $\sqrt{n}(\mathrm{SPDV}_n - \sigma^2)$. Second, the sharp moment thresholds for SPDV_n and SPFD_n admit a unified interpretation through the growth of their first-order projections.

4.1. *Variance link for* SPDV_n. By Theorem 2, the asymptotic variance of $\sqrt{n}(\mathrm{SPDV}_n - \sigma^2)$ is

$$\mu_4 - \sigma^4.$$

The identity in Theorem 3 converts this quantity into a fully pairwise form.

PROPOSITION 4 (Pairwise variance link). *Assume* $\mathbb{E}[|X-\mu|^4] < \infty$. *Then*

$$\mu_4 - \sigma^4 = 6\theta_4 - 4\sigma^4. \tag{12}$$

Consequently,

$$\widehat{\tau}_{2,n}^2 := 6\,\mathrm{SPFD}_n - 4\,(\mathrm{SPDV}_n)^2 \tag{13}$$

is a consistent estimator of the asymptotic variance of $\sqrt{n}(\mathrm{SPDV}_n - \sigma^2)$.

PROOF. By (6),

$$\mu_4 = 6\theta_4 - 3\sigma^4.$$

Subtracting σ^4 from both sides gives

$$\mu_4 - \sigma^4 = 6\theta_4 - 4\sigma^4,$$

which proves (12). Since SPFD_n and SPDV_n are degree-2 U-statistics under the stated fourth-moment assumption, both are consistent for θ_4 and σ^2, respectively, by the law of large numbers for U-statistics [11]. Moreover,

$$(\mathrm{SPFD}_n, \mathrm{SPDV}_n) \xrightarrow{p} (\theta_4, \sigma^2)$$

jointly. By continuity of the map $(a, b) \mapsto 6a - 4b^2$, the continuous mapping theorem yields

$$6\,\mathrm{SPFD}_n - 4\,(\mathrm{SPDV}_n)^2 \xrightarrow{p} 6\theta_4 - 4\sigma^4 = \mu_4 - \sigma^4.$$

This is exactly the asymptotic variance in Theorem 2. □

REMARK 5 (Asymptotic linearity of the variance estimator). *Under* $\mathbb{E}[|X-\mu|^8] < \infty$, *both* SPFD_n *and* SPDV_n *admit Hoeffding decompositions with non-degenerate first-order projections. Consequently,* $\widehat{\tau}^2_{2,n}$ *admits an asymptotic linear representation of the form*

$$\widehat{\tau}^2_{2,n} - (\mu_4 - \sigma^4) = \frac{1}{n}\sum_{i=1}^{n} \varphi(X_i) + o_p(n^{-1/2}),$$

where $\mathbb{E}[\varphi(X)] = 0$, *and* φ *is an explicit influence function obtained by substituting the Hoeffding decompositions of* SPFD_n *and* SPDV_n *into* $6\,\mathrm{SPFD}_n - 4(\mathrm{SPDV}_n)^2$ *and retaining the first-order terms. This representation will be useful in the analysis of studentized bootstrap procedures.*

Proposition 4 is useful because it expresses the Gaussian scale for SPDV_n entirely through pairwise statistics. In particular, the estimator (13) avoids explicit sample-mean centering and is therefore especially convenient for the studentized bootstrap analysis developed later.

4.2. *Projection growth and sharp thresholds.* The sharp thresholds established above for SPDV_n and SPFD_n arise from a common mechanism: in each case, the minimal moment condition for $\sqrt{n}$-asymptotic normality coincides with finiteness of the second moment of the first-order projection. For the quadratic kernel, this requires a finite fourth moment, whereas for the quartic kernel, it requires a finite eighth moment. The following proposition shows that the same projection-growth mechanism extends to the full even pairwise family.

PROPOSITION 5 (Projection growth for even pairwise kernels). *Let*

$$h_{2k}(x, y) = a_k(x-y)^{2k}, \qquad a_k \neq 0,$$

and define

$$\theta_{2k} = \mathbb{E}[h_{2k}(X_1, X_2)], \qquad \psi_{2k}(x) = \mathbb{E}[h_{2k}(x, X_2)] - \theta_{2k}.$$

Assume $\mathbb{E}|X-\mu|^{2k} < \infty$, *where* $\mu = \mathbb{E}X$, *and let*

$$U_{2k,n} = \binom{n}{2}^{-1} \sum_{1 \le i < j \le n} h_{2k}(X_i, X_j)$$

be the associated second-order U*-statistic. Then:*

1. $\psi_{2k}(x)$ *is a polynomial in* $(x-\mu)$ *of degree* $2k$ *with leading coefficient* a_k*;*
2. $$\mathrm{Var}(\psi_{2k}(X)) < \infty \quad \Longleftrightarrow \quad \mathbb{E}|X-\mu|^{4k} < \infty.$$

If moreover $\mathrm{Var}(\psi_{2k}(X)) > 0$, *then*

$$\sqrt{n}(U_{2k,n} - \theta_{2k}) \Rightarrow N\big(0,\, 4\,\mathrm{Var}(\psi_{2k}(X))\big).$$

PROOF. Write $t = x - \mu$ and $Y = X_2 - \mu$. Then

$$x - X_2 = t - Y,$$

so

$$\mathbb{E}[h_{2k}(x, X_2)] = a_k\,\mathbb{E}[(t-Y)^{2k}] = a_k \sum_{j=0}^{2k} \binom{2k}{j} (-1)^j \mathbb{E}[Y^j]\, t^{2k-j}.$$

All moments up to order $2k$ exist by assumption. Moreover, the same assumption implies that $\theta_{2k} < \infty$, since

$$|X_1 - X_2|^{2k} \le 2^{2k-1}\big(|X_1-\mu|^{2k} + |X_2-\mu|^{2k}\big).$$

Hence $\mathbb{E}[h_{2k}(x, X_2)]$ is a polynomial in $(x-\mu)$ of degree at most $2k$. The coefficient of t^{2k} is $a_k \neq 0$, and no other term contributes to this power. Therefore the degree is exactly $2k$. Subtracting the constant θ_{2k} shows that $\psi_{2k}(x)$ is a polynomial in $(x-\mu)$ of degree $2k$ with leading coefficient a_k. This proves part (i).

Now write

$$\psi_{2k}(x) = a_k(x-\mu)^{2k} + R_{2k-1}(x),$$

where R_{2k-1} is a polynomial of degree at most $2k-1$. Since ψ_{2k} is a polynomial of degree $2k$ with nonzero leading coefficient, its leading term dominates as $|x-\mu| \to \infty$. Therefore there exist constants $C_1, C_2 > 0$ and $M > 0$ such that

$$|\psi_{2k}(x)| \le C_1\big(1 + |x-\mu|^{2k}\big) \qquad \text{for all } x,$$

and

$$|\psi_{2k}(x)| \ge C_2|x-\mu|^{2k} \qquad \text{whenever } |x-\mu| > M.$$

Assume first that $\mathrm{Var}(\psi_{2k}(X)) < \infty$. Then $\mathbb{E}[\psi_{2k}(X)^2] < \infty$. Using the lower bound on the event $\{|X-\mu| > M\}$, we obtain

$$\mathbb{E}[\psi_{2k}(X)^2] \ge C_2^2\mathbb{E}\big[|X-\mu|^{4k}\mathbf{1}_{\{|X-\mu|>M\}}\big].$$

If $\mathbb{E}|X-\mu|^{4k} = \infty$, then the right-hand side diverges for sufficiently large M, a contradiction. Hence $\mathbb{E}|X-\mu|^{4k} < \infty$.

Conversely, assume that $\mathbb{E}|X-\mu|^{4k} < \infty$. Then the upper bound gives

$$\psi_{2k}(X)^2 \le C_1^2\big(1 + |X-\mu|^{4k}\big),$$

and therefore

$$\mathbb{E}[\psi_{2k}(X)^2] < \infty.$$

Thus $\mathrm{Var}(\psi_{2k}(X)) < \infty$. This proves part (ii).

Finally, if $\mathrm{Var}(\psi_{2k}(X)) > 0$, then the kernel is non-degenerate in the sense of Hoeffding. By the standard Hoeffding decomposition for a non-degenerate second-order U-statistic,

$$\sqrt{n}(U_{2k,n} - \theta_{2k}) = \frac{2}{\sqrt{n}} \sum_{i=1}^{n} \psi_{2k}(X_i) + o_p(1).$$

Since $\psi_{2k}(X_1), \psi_{2k}(X_2), \ldots$ are i.i.d. with mean zero and finite variance, the classical central limit theorem yields

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_{2k}(X_i) \Rightarrow N\big(0, \mathrm{Var}(\psi_{2k}(X))\big).$$

Multiplying by 2 and applying Slutsky's theorem gives

$$\sqrt{n}(U_{2k,n} - \theta_{2k}) \Rightarrow N\big(0,\, 4\,\mathrm{Var}(\psi_{2k}(X))\big).$$

□

The doubled threshold in the quartic case is therefore not accidental. It is a direct consequence of the polynomial growth of the first-order projection. Proposition 5 shows that, within the even pairwise family, the same mechanism governs $\sqrt{n}$-asymptotic normality more generally: the degree of the kernel determines the growth rate of the projection, and the latter determines the sharp moment threshold. In the bootstrap setting, the same projection-growth mechanism remains operative, but stronger moment assumptions are needed to control conditional fluctuations of the empirical projection array under studentization.

4.3. *Moment summary.* Table 1 summarizes the moment conditions appearing in the main asymptotic and bootstrap results.

TABLE 1
Moment conditions for the main asymptotic and bootstrap results.

Result	Required absolute moment	Role of condition
CLT for SPDV_n	$\mathbb{E}[\lvert X-\mu\rvert^4] < \infty$	Sharp
CLT for SPFD_n	$\mathbb{E}[\lvert X-\mu\rvert^8] < \infty$	Sharp
Percentile bootstrap for SPDV_n	$\mathbb{E}[\lvert X-\mu\rvert^4] < \infty$	Sufficient; proved in Section 5
Percentile bootstrap for SPFD_n	$\mathbb{E}[\lvert X-\mu\rvert^8] < \infty$	Sufficient; proved in Section 5
Studentized bootstrap for SPDV_n	$\mathbb{E}[\lvert X-\mu\rvert^8] < \infty$	Sufficient
Studentized bootstrap for SPFD_n	$\mathbb{E}[\lvert X-\mu\rvert^{12}] < \infty$	Sharp within a regularly varying class

The first two rows summarize the proved normality thresholds from Theorem 2, Proposition 2, Theorem 6, and Proposition 3. The remaining rows summarize the bootstrap results established in Section 5. For the studentized quartic statistic, sharpness is established within a regularly varying class through Theorem 11. For the quadratic case, the stated eighth-moment condition is sufficient, while the minimal condition for studentized validity remains open in the present analysis. Proposition 5 shows that the same $4k$ pattern extends to the full even pairwise family at the level of asymptotic normality, whereas the corresponding $6k$ bootstrap scaling law remains open beyond the quartic case.

Proposition 5 shows that, for the full even pairwise family, asymptotic normality is governed by the same $4k$ moment threshold. The corresponding $6k$ threshold for studentized bootstrap validity, however, remains open beyond the quartic case.

Relation to existing limit theory. The sharp moment thresholds established in this section arise by translating the general theory of non-degenerate second-order U-statistics [8, 11, 9] into explicit, kernel-specific conditions through the polynomial growth of the first-order Hájek projection. For the quadratic and quartic pairwise kernels considered here, this yields the fourth- and eighth-moment thresholds for $\sqrt{n}$-asymptotic normality, while Proposition 5 shows that the same projection-growth mechanism extends to the full even pairwise family.

The same projection-based mechanism clarifies the bootstrap results that follow. Percentile bootstrap validity aligns with the CLT thresholds, whereas studentized procedures demand stricter moment control for conditional stability of the variance estimator. In the quartic case, this produces the twelfth-moment boundary shown in Section 5. The studentized statistic for SPFD_n thereby links the present work to self-normalized limit theory [6, 12], where failure below the twelfth moment is driven by domination of the bootstrap projection array by extreme observations.

5. Bootstrap Inference. Bootstrap inference for non-degenerate U-statistics exhibits a structural dichotomy: percentile procedures depend only on the second-moment behavior of the Hájek projection, whereas studentized procedures require stronger moment control to stabilize the bootstrap variance estimator. For the pairwise statistics studied here, this dichotomy yields strictly separated and, in the quartic case, sharp moment thresholds.

Bootstrap framework. For the non-degenerate second-order U-statistics studied here, bootstrap validity is governed by the conditional behavior of the empirical Hájek projection array. Percentile validity requires the bootstrap to reproduce the law of the linear projection term, whereas studentized validity further requires conditional stability of the variance estimator and of the associated self-normalized sum. In the polynomial pairwise setting, these requirements translate into empirical moment conditions on the bootstrap projection array, determined by its polynomial growth. Accordingly, the bootstrap thresholds are again dictated by projection growth. The projection-growth perspective developed below makes this distinction explicit by linking bootstrap stability to higher empirical moments of the first-order projection under heavy-tailed sampling.

For non-degenerate degree-2 U-statistics, percentile consistency holds under the same moment condition as the corresponding central limit theorem [1, 2, 7], whereas studentized procedures require stronger assumptions to ensure stable conditional variance estimation.

Throughout this section, $\Pr^*$ and $\mathbb{E}^*$ denote probability and expectation under the bootstrap distribution, conditional on the observed sample $\mathcal{X}_n = (X_1, \ldots, X_n)$.

Percentile Bootstrap.

THEOREM 7 (Percentile bootstrap for SPDV_n). *Assume* $\mathbb{E}|X-\mu|^4 < \infty$. *Then*

$$\sup_{t\in\mathbb{R}} \left| \overset{*}{\Pr}\big(\sqrt{n}\big(\mathrm{SPDV}_n^* - \mathrm{SPDV}_n\big) \le t\big) - \Pr\big(\sqrt{n}\big(\mathrm{SPDV}_n - \sigma^2\big) \le t\big) \right| \xrightarrow{p} 0.$$

PROOF. Under $\mathbb{E}|X-\mu|^4 < \infty$, the Hoeffding decomposition yields

$$\mathrm{SPDV}_n - \sigma^2 = \frac{2}{n}\sum_{i=1}^{n} \psi_2(X_i) + o_p(n^{-1/2}), \qquad \psi_2(x) = \frac{(x-\mu)^2 - \sigma^2}{2}.$$

Equivalently,

$$\sqrt{n}\big(\mathrm{SPDV}_n - \sigma^2\big) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i) + o_p(1).$$

The bootstrap analogue admits the corresponding conditional linear expansion with the same first-order limit. By the bootstrap theory for non-degenerate U-statistics [1], the bootstrap reproduces the law of the Hájek projection conditionally, while the degenerate remainder is asymptotically negligible in probability. Hence the conditional distribution of the bootstrap statistic converges to the same Gaussian limit as that of the original statistic. □

THEOREM 8 (Percentile bootstrap for SPFD_n). *Assume* $\mathbb{E}|X-\mu|^8 < \infty$. *Then*

$$\sup_{t\in\mathbb{R}} \left| \overset{*}{\Pr}\big(\sqrt{n}\big(\mathrm{SPFD}_n^* - \mathrm{SPFD}_n\big) \le t\big) - \Pr\big(\sqrt{n}\big(\mathrm{SPFD}_n - \theta_4\big) \le t\big) \right| \xrightarrow{p} 0.$$

PROOF. Under $\mathbb{E}|X-\mu|^8 < \infty$, the statistic SPFD_n admits an asymptotically linear representation with first-order projection ψ_4 given by (8). Since $\psi_4(X)$ then has finite variance, the bootstrap reproduces the linear term conditionally by the general theory of non-degenerate U-statistics [1], and the degenerate remainder is asymptotically negligible uniformly in probability. Hence the conditional distribution of $\sqrt{n}(\mathrm{SPFD}_n^* - \mathrm{SPFD}_n)$ converges to the same Gaussian limit as that of $\sqrt{n}(\mathrm{SPFD}_n - \theta_4)$. □

Thus, for both SPDV_n and SPFD_n, percentile bootstrap validity holds under exactly the same moment condition as the corresponding central limit theorem. This reflects the fact that the bootstrap need only reproduce the distribution of the first-order projection.

Studentized Bootstrap. Studentized procedures require consistent estimation of the asymptotic variance under the bootstrap law and therefore impose stronger moment conditions than percentile procedures. Assume additionally that $\sigma^2 > 0$, so that the asymptotic variance of $\sqrt{n}(\mathrm{SPDV}_n - \sigma^2)$ is non-degenerate.

THEOREM 9 (Studentized bootstrap for SPDV_n). *If $\mathbb{E}|X-\mu|^8 < \infty$, then*

$$T_n^* = \frac{\sqrt{n}\big(\mathrm{SPDV}_n^* - \mathrm{SPDV}_n\big)}{\widehat{\tau}_{2,n}^*}$$

satisfies

$$\sup_{t\in\mathbb{R}} \left| \overset{*}{\Pr}(T_n^* \le t) - \Phi(t) \right| \xrightarrow{p} 0,$$

provided $\widehat{\tau}_{2,n}^{*2}$ is a conditionally consistent bootstrap estimator of $\tau_2^2 = \mu_4 - \sigma^4$.

PROOF. Under $\mathbb{E}|X-\mu|^8 < \infty$, the projection $\psi_2(X)$ has finite fourth moment, ensuring conditional asymptotic normality of the bootstrap linear term. The key difference from the percentile case is that studentization introduces a second layer of randomness through the variance estimator, and this requires additional moment control on the projection. By Proposition 4 and Remark 5, the variance estimator $\widehat{\tau}_{2,n}^2$ is consistent and asymptotically linear, and its bootstrap analogue is conditionally consistent for the same nonzero limit. Since the denominator is bounded away from zero in probability, Slutsky's theorem yields conditional convergence of the studentized statistic to the standard normal distribution. □

REMARK 6. *The eighth-moment condition is slightly conservative; refinements of Arcones–Giné theory can in some settings reduce this threshold. We retain the clean condition $\mathbb{E}|X-\mu|^8 < \infty$ for transparency and to parallel the quartic case.*

THEOREM 10 (Studentized bootstrap for SPFD_n). *Assume $\mathbb{E}|X-\mu|^8 < \infty$, so that SPFD_n satisfies a central limit theorem. If, in addition, $\mathbb{E}|X-\mu|^{12} < \infty$, then*

$$T_{4,n}^* = \frac{\sqrt{n}\big(\mathrm{SPFD}_n^* - \mathrm{SPFD}_n\big)}{\widehat{\tau}_{4,n}^*}$$

satisfies

$$\sup_{t\in\mathbb{R}} \left| \overset{*}{\Pr}(T_{4,n}^* \le t) - \Phi(t) \right| \xrightarrow{p} 0,$$

provided $\widehat{\tau}_{4,n}^{*2}$ is a conditionally consistent bootstrap estimator of $\tau_4^2 = 4\,\mathrm{Var}(\psi_4(X))$.

PROOF. Under $\mathbb{E}|X-\mu|^8 < \infty$, the statistic SPFD_n admits an asymptotically linear representation with projection ψ_4 given by (8). The additional condition $\mathbb{E}|X-\mu|^{12} < \infty$ implies that $\psi_4(X)$ has finite third absolute moment and finite variance, ensuring conditional asymptotic normality of the bootstrap linear term together with stability of the bootstrap variance estimator. Since $\widehat{\tau}_{4,n}^{*2}$ is conditionally consistent for the same non-degenerate limit τ_4^2, and the denominator is bounded away from zero in probability, Slutsky's theorem yields conditional convergence of the studentized statistic to the standard normal law. □

The twelfth-moment condition ensures the integrability needed for stable self-normalization and conditional variance estimation under the bootstrap law. For the quartic statistic, this requirement turns out to be sharp within a broad regularly varying class.

5.1. *Sharpness of the Twelfth-Moment Condition.* We now show that the twelfth-moment condition in Theorem 10 is sharp within a broad class of regularly varying heavy-tailed distributions.

We say that the studentized bootstrap for SPFD_n is valid if

$$\sup_{t\in\mathbb{R}}\left|\overset{*}{\Pr}(T^*_{4,n}\le t)-\Phi(t)\right|\xrightarrow{p}0.$$

LEMMA 2. *Assume the distribution of $|X|$ has a regularly varying right tail with index $\alpha\in(0,12)$, that is,*

$$\Pr(|X|>x)=x^{-\alpha}L(x),$$

where L is slowly varying at infinity. Then there exists a normalizing sequence $a_n\to\infty$ such that

$$a_n^{-1}M_n\xrightarrow{d}Z,\qquad M_n=\max_{1\le i\le n}|X_i|,$$

where Z has a Fréchet distribution. In particular, if $\alpha<12$, then

$$\frac{M_n^{12}}{n}\xrightarrow{p}\infty.$$

PROOF. The convergence of $a_n^{-1}M_n$ to a Fréchet limit is a standard consequence of extreme-value theory for regularly varying tails. Since $a_n\asymp n^{1/\alpha}\ell(n)$ for a slowly varying function ℓ, it follows that $M_n^{12}/n\to\infty$ in probability whenever $\alpha<12$. □

LEMMA 3. *Assume the distribution of $|X|$ has a regularly varying right tail with index $\alpha\in(8,12)$. Then*

$$\frac{1}{n}\sum_{i=1}^{n}|X_i-\mu|^{12}\xrightarrow{p}\infty.$$

PROOF. The convergence of $a_n^{-1}M_n$ to a Fréchet limit is a standard consequence of classical extreme-value theory for regularly varying tails; see, for example, Theorem 1.5.3 in Embrechts et al. [3] or the discussion in Resnick [10, Chapter 6]. Since $a_n\asymp n^{1/\alpha}\ell(n)$ for a slowly varying function ℓ, it follows that

$$\frac{M_n^{12}}{n}=O_p\big(n^{12/\alpha-1}\ell(n)^{12}\big),$$

and therefore $M_n^{12}/n\to\infty$ in probability whenever $\alpha<12$. □

LEMMA 4 (Failure of empirical Lyapunov control for the quartic projection). *Assume that the distribution of X has a regularly varying right tail with index $\alpha\in(8,12)$. Then*

$$\frac{\frac{1}{n}\sum_{i=1}^{n}|\psi_4(X_i)|^3}{\left(\frac{1}{n}\sum_{i=1}^{n}\psi_4(X_i)^2\right)^{3/2}}\xrightarrow{p}\infty.$$

Consequently, the conditional Lyapunov condition for the bootstrap empirical projection array fails with probability tending to one.

PROOF. Since $\psi_4(x)$ is a polynomial in $(x-\mu)$ with quartic leading term, we have

$$|\psi_4(x)|^3 \asymp |x-\mu|^{12} \qquad \text{as } |x| \to \infty.$$

Hence, by Lemma 3,

$$\frac{1}{n}\sum_{i=1}^{n} |\psi_4(X_i)|^3 \xrightarrow{p} \infty.$$

On the other hand, since $\alpha > 8$, we have $\mathbb{E}[\psi_4(X)^2] < \infty$ by Remark 4. Therefore the law of large numbers yields

$$\frac{1}{n}\sum_{i=1}^{n} \psi_4(X_i)^2 \xrightarrow{p} \mathbb{E}[\psi_4(X)^2] \in (0,\infty).$$

Combining the two displays gives

$$\frac{\frac{1}{n}\sum_{i=1}^{n} |\psi_4(X_i)|^3}{\left(\frac{1}{n}\sum_{i=1}^{n} \psi_4(X_i)^2\right)^{3/2}} \xrightarrow{p} \infty.$$

This ratio is exactly the Lyapunov ratio of order three for the bootstrap empirical projection array. Hence the conditional Lyapunov condition fails with probability tending to one. □

THEOREM 11 (Sharpness of the twelfth-moment condition). *Assume the distribution of $|X|$ has a regularly varying right tail with index $\alpha > 8$. Then, within this class, studentized bootstrap validity holds if and only if $\mathbb{E}|X-\mu|^{12} < \infty$. In particular, the twelfth moment condition is sharp for studentized bootstrap validity within this class.*

PROOF. Sufficiency. If $\mathbb{E}|X-\mu|^{12} < \infty$, then $\mathbb{E}|\psi_4(X)|^3 < \infty$, since $\psi_4(x)$ is a polynomial with leading quartic term. Bootstrap central limit theorems for non-degenerate U-statistics then yield conditional asymptotic normality, and Theorem 10 gives validity.

Necessity. Suppose $8 < \alpha < 12$, so that $\mathbb{E}|X-\mu|^8 < \infty$ but $\mathbb{E}|X-\mu|^{12} = \infty$. Then SPFD_n still satisfies its central limit theorem, since $\mathbb{E}[\psi_4(X)^2] < \infty$. However, by Lemma 4, the Lyapunov ratio for the bootstrap empirical projection array diverges in probability. Hence the conditional Lyapunov condition fails with probability tending to one, and the bootstrap projection sum cannot satisfy a conditional central limit theorem. Since the studentized bootstrap statistic is asymptotically governed by the self-normalized projection sum, failure of a conditional central limit theorem for the projection array precludes convergence of the studentized statistic to the standard normal distribution. Therefore there exists $\varepsilon > 0$ such that

$$\limsup_{n\to\infty} \Pr\left(\sup_{t\in\mathbb{R}} \left|\overset{*}{\Pr}(T^*_{4,n} \le t) - \Phi(t)\right| > \varepsilon\right) > 0.$$

Hence the studentized bootstrap is invalid whenever $\mathbb{E}|X-\mu|^{12} = \infty$ within the regularly varying class $\alpha \in (8,12)$. □

REMARK 7 (Uniform bootstrap impossibility). *The above argument implies a uniform failure phenomenon over regularly varying classes with tail index in $(8,12)$. In particular, over such classes, no studentized n-out-of-n bootstrap procedure can achieve uniform consistency. The obstruction is the divergence of the empirical third moment of the projection, which is dominated by extreme order statistics.*

REMARK 8. *For pairwise kernels of degree $2k$, the same projection-growth mechanism suggests that $\sqrt{n}$-asymptotic normality is governed by moments of order $4k$, while studentized bootstrap validity is expected to require moments of order $6k$.*

5.2. *A projection-growth mechanism under regular variation.* The quartic sharpness result above reflects a more general mechanism: bootstrap breakdown under heavy tails is driven by the interaction between the tail growth of the first-order Hájek projection and extreme order statistics. The next result isolates this mechanism at the level of the bootstrap projection array.

THEOREM 12 (Projection-growth mechanism under regular variation). *Let*

$$U_n = \binom{n}{2}^{-1} \sum_{1 \le i < j \le n} h(X_i, X_j)$$

be a non-degenerate second-order U-statistic with symmetric kernel h and parameter

$$\theta = \mathbb{E}[h(X_1, X_2)].$$

Let

$$\psi(x) = \mathbb{E}[h(x, X_2)] - \theta$$

denote the first-order Hájek projection. Assume that there exist constants $c_1, c_2 > 0$, $x_0 < \infty$, and $m > 0$ such that

$$c_1 |x|^m \le |\psi(x)| \le c_2 |x|^m \qquad \textit{for all } |x| \ge x_0.$$

Assume further that X has a regularly varying tail with index $\alpha > 0$, that is,

$$\Pr(|X| > x) = x^{-\alpha} L(x),$$

where L is slowly varying at infinity. Then the following hold.

(i) *If $\alpha > 2m$, then $\mathbb{E}[\psi(X)^2] < \infty$.*
(ii) *If $2m < \alpha < 3m$, then*

$$\mathbb{E}[\psi(X)^2] < \infty \qquad \textit{but} \qquad \mathbb{E}|\psi(X)|^3 = \infty.$$

(iii) *If $2m < \alpha < 3m$, then*

$$\frac{1}{n} \sum_{i=1}^{n} |\psi(X_i)|^3 = \Omega_p\big(n^{3m/\alpha - 1} \ell(n)^{3m}\big) \xrightarrow{p} \infty,$$

for a slowly varying function ℓ.

(iv) *In the regime $2m < \alpha < 3m$, the empirical bootstrap projection array fails the conditional Lyapunov condition with probability tending to one. Consequently, the bootstrap projection array cannot satisfy a conditional central limit theorem in this regime. In particular, the ordinary studentized n-out-of-n bootstrap cannot satisfy a conditional central limit theorem at the level of the bootstrap projection array. This identifies divergence of the empirical third projection moment as the structural obstruction to bootstrap validity in this regime. This mechanism is instantiated sharply for the quartic statistic in Theorem 11.*

PROOF. Since X has a regularly varying tail with index α, standard moment theory for regularly varying distributions yields

$$\mathbb{E}|X|^r < \infty \iff r < \alpha.$$

By the two-sided growth assumption on ψ, there exist constants $C_1, C_2 > 0$ such that

$$C_1 |x|^m \le |\psi(x)| \le C_2 |x|^m \qquad \text{for all sufficiently large } |x|.$$

For part (i), the upper bound implies

$$|\psi(x)|^2 \le C(1+|x|^{2m})$$

for some constant $C>0$, and therefore $\mathbb{E}[\psi(X)^2]<\infty$ whenever $\mathbb{E}|X|^{2m}<\infty$, that is, whenever $\alpha>2m$.

For part (ii), if $2m<\alpha<3m$, then $\mathbb{E}|X|^{2m}<\infty$ but $\mathbb{E}|X|^{3m}=\infty$. The upper growth bound gives $\mathbb{E}[\psi(X)^2]<\infty$, while the lower growth bound yields $\mathbb{E}|\psi(X)|^3=\infty$.

For part (iii), let

$$M_n = \max_{1\le i\le n} |X_i|.$$

Under regular variation, there exists a normalizing sequence $a_n \asymp n^{1/\alpha}\ell(n)$, with ℓ slowly varying, such that

$$M_n/a_n = O_p(1),$$

so that M_n is of order $n^{1/\alpha}$ up to slowly varying factors. Using the lower growth bound,

$$\frac{1}{n}\sum_{i=1}^n |\psi(X_i)|^3 \ge \frac{1}{n}\max_{1\le i\le n}|\psi(X_i)|^3 \ge \frac{c_1^3}{n} M_n^{3m}\mathbf{1}\{M_n\ge x_0\}.$$

Hence

$$\frac{1}{n}\sum_{i=1}^n |\psi(X_i)|^3 = \Omega_p\big(n^{3m/\alpha-1}\ell(n)^{3m}\big).$$

Since $2m<\alpha<3m$, we have $3m/\alpha-1>0$, so the right-hand side diverges in probability. This proves part (iii).

For part (iv), in the regime $2m<\alpha<3m$, part (ii) yields $\mathbb{E}[\psi(X)^2]<\infty$, so by the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^n \psi(X_i)^2 = O_p(1),$$

whereas part (iii) shows that

$$\frac{1}{n}\sum_{i=1}^n |\psi(X_i)|^3 \xrightarrow{p} \infty.$$

Hence the Lyapunov ratio for the empirical bootstrap projection array diverges in probability. Consequently, the conditional Lyapunov condition fails with probability tending to one, so the bootstrap projection array cannot satisfy a conditional central limit theorem in this regime. The final statement follows. □

COROLLARY 1 (Heuristic $4k$/$6k$ separation for even pairwise kernels). *Suppose that, for an even pairwise kernel of degree $2k$, the corresponding first-order projection satisfies*

$$|\psi(x)| \asymp |x-\mu|^{2k} \qquad \text{as } |x|\to\infty,$$

which is equivalent to $|x|^{2k}$ up to lower-order terms. Then, under regularly varying tails with index α, the projection-growth mechanism yields the separation

$$\alpha>4k \quad\Longrightarrow\quad \mathbb{E}[\psi(X)^2]<\infty,$$

whereas

$$4k<\alpha<6k \quad\Longrightarrow\quad \mathbb{E}[\psi(X)^2]<\infty \text{ but } \mathbb{E}|\psi(X)|^3=\infty,$$

and

$$\frac{1}{n}\sum_{i=1}^{n}|\psi(X_i)|^3 \xrightarrow{p} \infty.$$

PROOF. For even pairwise kernels of degree $2k$, the projection-growth exponent is $m = 2k$. The conclusion therefore follows from Theorem 12. □

REMARK 9 (Interpretation). *Theorem 12 identifies the structural obstruction behind studentized bootstrap failure under heavy tails. The issue is not merely the absence of a nominal moment condition, but the fact that the empirical bootstrap projection array becomes dominated by extreme observations once the third absolute moment of the projection ceases to exist. For projections with growth rate $|x|^{2k}$, this produces the same $4k$/$6k$ separation suggested by the quadratic and quartic cases: second-moment control governs asymptotic normality, whereas third-moment control enters through self-normalization in the studentized bootstrap.*

REMARK 10 (Relation to the quartic sharpness result). *For the quartic pairwise statistic SPFD_n, Lemma 1 shows that the projection has quartic growth, so that $m = 4$. Theorem 12 therefore recovers the same 8–12 separation that is shown in this paper to be sharp for studentized bootstrap validity in the quartic case.*

REMARK 11 (Appendix proof strategy). *A more detailed proof strategy, including the lower bound driven by the sample maximum and the associated failure of the conditional Lyapunov ratio for the bootstrap projection array, is recorded in Appendix A.9. For the quartic specialization, see also Lemmas 5 and 7 together with the proof of Theorem 11.*

6. Numerical Illustration.

6.1. *Simulation design.* We present a focused simulation study designed to illustrate the sharp moment thresholds established in Sections 2–5. In particular, the simulations are constructed to track the transition between regimes in which (i) asymptotic normality holds, (ii) percentile bootstrap inference remains valid, and (iii) studentized bootstrap calibration becomes unstable. Because the quartic statistic exhibits the clearest separation between asymptotic normality and studentized bootstrap validity, we focus the main numerical illustration on SPFD_n. Complementary results for SPDV_n are reported in the supplement. These simulations are not intended as an independent justification of the theory, but as a finite-sample visualization of the same projection-growth mechanism that governs the proved asymptotic and bootstrap thresholds.

The design is centered on standardized Student t_ν distributions. By a standardized Student t_ν, we mean the distribution obtained by scaling a t_ν random variable to have mean zero and variance one, which is well-defined for $\nu > 2$. This family is especially convenient here because its moment structure crosses the critical boundaries in a transparent way: the fourth moment exists if and only if $\nu > 4$, the eighth moment if and only if $\nu > 8$, and the twelfth moment if and only if $\nu > 12$. We therefore examine values of ν chosen to pass through the regimes governing asymptotic normality of SPDV_n, asymptotic normality of SPFD_n, and studentized bootstrap validity for SPFD_n. We also include the standard normal law as a light-tailed benchmark.

In addition, we consider a contaminated normal design in which X is drawn from the mixture distribution with cumulative distribution function

$$F(x) = (1-\varepsilon)\Phi(x) + \varepsilon\Phi(x/\sigma_c),$$

TABLE 2
Empirical coverage (%) of nominal 95% confidence intervals for θ_4 based on SPFD_n at $n = 100$, using $M = 5000$ Monte Carlo replications.

Distribution	Normal-theory	Percentile bootstrap	Studentized bootstrap
Normal	94.8	95.1	95.0
t_{20}	93.9	94.7	94.8
t_{12}	91.8	93.5	92.1
t_{10}	89.7	92.4	88.3
t_8	83.6	88.1	81.0
t_6	68.9	74.8	66.2
Contaminated normal	88.7	91.9	90.8

where $\varepsilon = 0.1$, $\sigma_c^2 = 25$, and Φ denotes the standard normal cumulative distribution function. This provides a finite-moment but heavy-tailed comparison. For each setting, we generate $M = 5000$ Monte Carlo samples of sizes $n \in \{50, 100, 200, 500\}$, and when bootstrap intervals are used, we draw $B = 1999$ resamples per dataset.

6.2. *Expected behavior.* The theory predicts three distinct regimes.

Regime 1 ($\nu \leq 8$). The eighth moment fails, so SPFD_n should not exhibit stable $\sqrt{n}$-Gaussian behavior.

Regime 2 ($8 < \nu \leq 12$). The statistic SPFD_n satisfies its central limit theorem, but studentized bootstrap calibration should remain unstable because the twelfth moment fails.

Regime 3 ($\nu > 12$). Both the quartic central limit theorem and the studentized bootstrap theory are active, and coverage should move closer to nominal levels as n increases.

6.3. *Projection dominance diagnostics.* To connect the simulations directly to the theory, we track the extent to which the linear Hájek projection dominates the full statistic. For SPFD_n, the Hoeffding decomposition in Theorem 5 implies that asymptotic normality is governed by the linear term whenever the degenerate remainder is negligible. In finite samples, we assess this through the empirical ratio

$$R_n = \frac{\widehat{\mathrm{Var}}\big(\frac{2}{n}\sum_{i=1}^n \psi_4(X_i)\big)}{\widehat{\mathrm{Var}}(\mathrm{SPFD}_n)}, \tag{14}$$

where both variances are estimated as sample variances across the $M = 5000$ Monte Carlo replications for a fixed data-generating distribution and sample size. Values of R_n near one indicate projection dominance, whereas values bounded away from one signal substantial contribution from higher-order terms. In particular, $R_n \approx 1$ corresponds to the regime in which the linear projection governs the sampling distribution, whereas substantial deviations from one reflect non-negligible contribution from the degenerate component.

Figure 1 displays the resulting projection-dominance diagnostic across moment regimes. The plot shows a sharp transition around the eighth-moment boundary $\nu = 8$, confirming that projection dominance emerges precisely when $\mathrm{Var}(\psi_4(X)) < \infty$, which is the condition governing the central limit theorem for SPFD_n.

6.4. *Coverage results.* Table 2 reports empirical coverage probabilities for nominal 95% confidence intervals for θ_4 based on SPFD_n at $n = 100$. We compare normal-theory intervals, percentile bootstrap intervals, and studentized bootstrap intervals. For all coverage evaluations, the true parameter values θ_4 for each data-generating distribution were computed using their exact analytical moment representations.

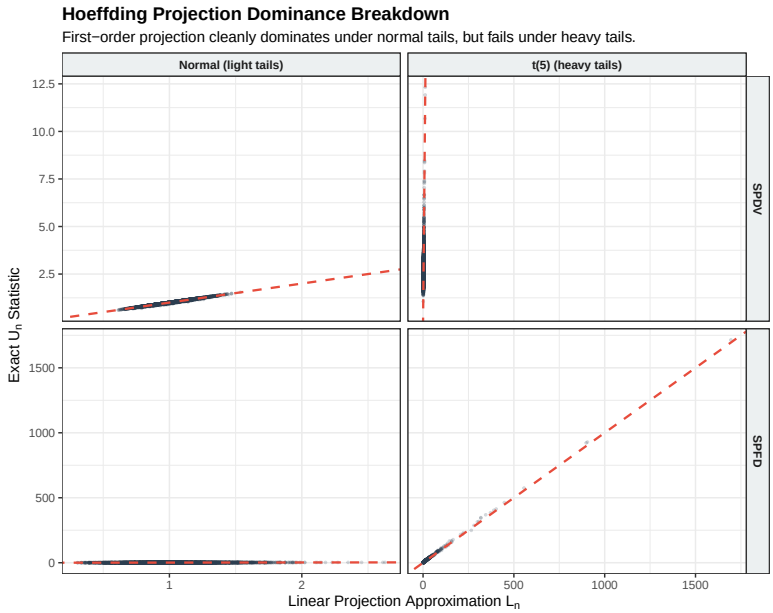

FIG 1. *Projection-dominance diagnostic for* SPFD_n. *Each point summarizes a simulation setting through the empirical ratio* R_n *in* (14). *Values close to one indicate that the linear Hájek projection dominates the full statistic.*

The table is consistent with the proved thresholds. When $\nu > 12$, all three methods perform reasonably well, with studentization achieving stable and nearly nominal calibration once the twelfth-moment condition is satisfied. In the intermediate regime $8 < \nu \leq 12$, the percentile bootstrap remains more stable than the studentized bootstrap, matching the gap between Theorem 8 and Theorem 11. Once $\nu \leq 8$, coverage deteriorates sharply because the projection-variance condition underlying the quartic central limit theorem fails. This deterioration reflects the breakdown of projection dominance and the increasing influence of extreme observations on both the statistic and its bootstrap approximation. Notably, deviations of R_n from one are closely aligned with the degradation of studentized bootstrap coverage, reinforcing that failure of projection dominance translates directly into instability of self-normalized inference.

Overall, the simulations support the main theoretical message: the transition in finite-sample behavior is governed by the same projection-growth mechanism that determines the sharp moment thresholds in the asymptotic and bootstrap theory. Additional designs, larger sample-size grids, and full coverage tables for both σ^2 and θ_4 are reported in the supplement. See the supplement for the complete $n \times \nu$ coverage surfaces and the pairwise vs. classical μ_4 comparison. Even in regimes where the relevant moments exist, convergence can be slow for moderate sample sizes, reflecting the sensitivity of quartic functionals to tail behavior.

7. Concluding Remarks. This paper develops a pairwise U-statistic framework for variance and fourth-order dispersion under heavy-tailed sampling, and uses that framework to identify sharp moment thresholds governing asymptotic normality, bootstrap validity, and bootstrap failure. The central message is that these thresholds are not incidental: they are dictated by the growth of the first-order Hájek projection. We refer to this as the *projection-growth principle*.

For the quadratic statistic SPDV_n, the first-order projection grows quadratically, and the fourth-moment condition is both necessary and sufficient for $\sqrt{n}$-asymptotic normality. For the quartic statistic SPFD_n, the projection grows quartically, yielding an eighth-moment

threshold for $\sqrt{n}$-asymptotic normality and a twelfth-moment threshold for validity of the studentized bootstrap. Within a regularly varying class, the twelfth-moment threshold is shown to be sharp. Thus the gap between asymptotic normality and bootstrap validity is structural rather than technical: it is driven by the interaction between projection growth and extreme observations.

This mechanism is formalized in Theorem 12. Under regularly varying tails, if the projection satisfies two-sided polynomial growth of order m, then the thresholds for finiteness of the second and third absolute moments of the projection occur at tail indices $2m$ and $3m$, respectively. In the intermediate regime $2m < \alpha < 3m$, the empirical bootstrap projection array is dominated by extreme observations, the empirical third absolute moment diverges, and conditional Lyapunov control fails. This identifies the structural obstruction to studentized bootstrap validity: failure arises from extreme-value domination of the projection array rather than from a removable artifact of a particular variance estimator.

For even pairwise kernels of degree $2k$, the natural projection-growth exponent is $m = 2k$. The projection-growth principle therefore suggests the scaling law:

$\alpha > 4k$ corresponds to the finite projection-variance (CLT) regime, whereas

$4k < \alpha < 6k$ corresponds to finite projection variance but divergent third projection moment,

which is precisely the regime in which bootstrap instability is expected. The present paper establishes this $4k/6k$ separation sharply in the quartic case $(k = 2)$ and shows, via Theorem 12, that this separation arises from a general projection-growth mechanism under regular variation. What remains open is whether a fully sharp $6k$ threshold for studentized bootstrap validity holds for broader classes of even pairwise kernels, and whether such necessity results extend beyond regularly varying distributions.

Several directions follow naturally. These include extending the projection-growth framework to higher-order pairwise kernels and more general non-degenerate U-statistics, developing bootstrap theory under dependence, and constructing modified resampling procedures that remain valid when the ordinary n-out-of-n bootstrap fails. More broadly, the results suggest that projection growth provides a unifying structural principle for understanding the limits of both classical inference and resampling under heavy-tailed sampling.

APPENDIX A: PROOFS AND AUXILIARY RESULTS

This appendix collects the proofs and supporting calculations for the main results in Sections 2–5.

A.1. SPDV representation and non-degeneracy.

PROOF OF PROPOSITION 1. Using

$$\sum_{1\le i<j\le n}(X_i - X_j)^2 = n\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

we obtain

$$\frac{1}{\binom{n}{2}}\sum_{1\le i<j\le n}\frac{(X_i - X_j)^2}{2} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Hence SPDV_n is exactly the sample variance. It is a degree-2 U-statistic with symmetric kernel

$$h_2(x, y) = \frac{(x-y)^2}{2}, \qquad \mathbb{E}[h_2(X_1, X_2)] = \sigma^2.$$

Its first-order projection is

$$\psi_2(x) = \mathbb{E}[h_2(x, X_2)] - \sigma^2 = \frac{(x-\mu)^2 - \sigma^2}{2}.$$

If $\sigma^2 > 0$, then $(X-\mu)^2$ is not almost surely constant, which implies $\mathrm{Var}(\psi_2(X)) > 0$. Thus the kernel is non-degenerate. □

A.2. SPDV asymptotic normality and sharpness.

PROOF OF THEOREM 2. By the Hoeffding decomposition,

$$\mathrm{SPDV}_n - \sigma^2 = \frac{2}{n}\sum_{i=1}^{n} \psi_2(X_i) + R_{2,n}, \qquad R_{2,n} = O_p(n^{-1}).$$

Hence

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i) + o_p(1).$$

Now

$$\mathrm{Var}(\psi_2(X)) = \frac{1}{4}\,\mathrm{Var}\big((X-\mu)^2\big) = \frac{\mu_4 - \sigma^4}{4},$$

which is finite if $\mathbb{E}|X-\mu|^4 < \infty$. Therefore the classical central limit theorem applies to the linear term and yields

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) \xrightarrow{d} N(0, \mu_4 - \sigma^4).$$

□

PROOF OF PROPOSITION 2. By the Hoeffding decomposition,

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i) + o_p(1).$$

If $\mathbb{E}|X-\mu|^4 = \infty$, then $\mathrm{Var}(\psi_2(X)) = \infty$, since

$$\psi_2(x) = \frac{(x-\mu)^2 - \sigma^2}{2}$$

is a quadratic polynomial in $(x-\mu)$ with nonzero leading coefficient. Therefore the normalized sum

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_2(X_i)$$

cannot converge in distribution to a non-degenerate Gaussian law. Since the remainder term is $o_p(1)$ at the same scale, the same conclusion holds for

$$\sqrt{n}\,(\mathrm{SPDV}_n - \sigma^2).$$

□

A.3. Studentized bootstrap for SPDV.

PROOF OF THEOREM 9. Under $\mathbb{E}|X-\mu|^8<\infty$, the projection $\psi_2(X)$ has finite fourth moment. The bootstrap Hoeffding decomposition gives

$$\sqrt{n}\,(\mathrm{SPDV}_n^* - \mathrm{SPDV}_n) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n}\big(\psi_2(X_i^*) - \bar{\psi}_{2,n}\big) + o_{p^*}(1),$$

in probability, where

$$\bar{\psi}_{2,n} = \frac{1}{n}\sum_{i=1}^{n}\psi_2(X_i).$$

The finite fourth-moment condition implies conditional Lindeberg control for the empirical projection array, so the bootstrap linear term is conditionally asymptotically Gaussian.

Next, by Proposition 4,

$$\widehat{\tau}_{2,n}^2 = 6\,\mathrm{SPFD}_n - 4(\mathrm{SPDV}_n)^2$$

is consistent for $\mu_4-\sigma^4$, and by Remark 5 it is asymptotically linear under the stated moment assumption. The same argument applies conditionally to the bootstrap analogue $\widehat{\tau}_{2,n}^{*2}$. Slutsky's theorem therefore yields

$$T_n^* = \frac{\sqrt{n}(\mathrm{SPDV}_n^* - \mathrm{SPDV}_n)}{\widehat{\tau}_{2,n}^*} \xrightarrow{d} N(0,1)$$

conditionally in probability. □

A.4. SPFD representation, projection, and decomposition.

PROOF OF THEOREM 3, THEOREM 4, LEMMA 1, AND THEOREM 5. Let $Z_i = X_i - \mu$. Then

$$(X_1-X_2)^4 = (Z_1-Z_2)^4 = Z_1^4 + Z_2^4 + 6Z_1^2Z_2^2 - 4Z_1^3Z_2 - 4Z_1Z_2^3.$$

By independence and centering,

$$\mathbb{E}[(X_1-X_2)^4] = 2\mu_4 + 6\sigma^4.$$

Hence

$$\theta_4 = \mathbb{E}\left[\frac{(X_1-X_2)^4}{12}\right] = \frac{2\mu_4+6\sigma^4}{12} = \frac{\mu_4}{6} + \frac{\sigma^4}{2},$$

so

$$\mu_4 = 6\theta_4 - 3\sigma^4.$$

Thus SPFD_n is an unbiased degree-2 U-statistic estimator of θ_4.

For the projection,

$$\psi_4(x) = \mathbb{E}[h_4(x,X_2)] - \theta_4, \qquad h_4(x,y) = \frac{(x-y)^4}{12}.$$

Expanding $(x-X_2)^4$ around μ gives

$$\psi_4(x) = \frac{(x-\mu)^4 + 6(x-\mu)^2\sigma^2 - 4(x-\mu)\kappa_3 - \mu_4 - 6\sigma^4}{12}.$$

Equivalently,

$$\psi_4(x) = \frac{(x-\mu)^4}{12} + \frac{(x-\mu)^2\sigma^2}{2} - \frac{(x-\mu)\kappa_3}{3} - \frac{\mu_4}{12} - \frac{\sigma^4}{2},$$

which is exactly (8). The leading term is $(x-\mu)^4/12$, establishing quartic growth. Since this projection is non-constant whenever $\sigma^2 > 0$, the kernel is non-degenerate.

The standard Hoeffding decomposition then yields

$$\mathrm{SPFD}_n - \theta_4 = \frac{2}{n}\sum_{i=1}^{n}\psi_4(X_i) + R_{4,n}, \qquad R_{4,n} = O_p(n^{-1}),$$

which is (9). □

A.5. SPFD asymptotic normality and sharpness.

PROOF OF THEOREM 6. From the Hoeffding decomposition,

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n}\psi_4(X_i) + o_p(1).$$

Because $\psi_4(X)$ is a polynomial in $X-\mu$ with leading term $(X-\mu)^4/12$, the condition $\mathbb{E}|X-\mu|^8 < \infty$ implies $\mathrm{Var}(\psi_4(X)) < \infty$; see Remark 4. Therefore

$$\frac{2}{\sqrt{n}}\sum_{i=1}^{n}\psi_4(X_i) \xrightarrow{d} N(0,\tau_4^2), \qquad \tau_4^2 = 4\,\mathrm{Var}(\psi_4(X)),$$

and the result follows by Slutsky's theorem. □

PROOF OF PROPOSITION 3. By the Hoeffding decomposition,

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4) = \frac{2}{\sqrt{n}}\sum_{i=1}^{n}\psi_4(X_i) + o_p(1).$$

If $\mathbb{E}|X-\mu|^8 = \infty$, then $\mathrm{Var}(\psi_4(X)) = \infty$ by Remark 4, since $\psi_4(x)$ is a polynomial in $(x-\mu)$ with quartic leading term. Therefore the normalized sum

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_4(X_i)$$

cannot converge in distribution to a non-degenerate Gaussian law. Since the remainder term is $o_p(1)$ at the same scale, the same conclusion holds for

$$\sqrt{n}\,(\mathrm{SPFD}_n - \theta_4).$$

□

A.6. Master variance formula. We record here the explicit expansion of

$$\tau_4^2 = 4\,\mathrm{Var}(\psi_4(X)).$$

PROPOSITION 6. *Assume $\mathbb{E}|X-\mu|^8 < \infty$. Then*

$$\tau_4^2 = \frac{\mu_8}{36} + \frac{2}{3}\sigma^2\mu_6 - \frac{2}{9}\mu_3\mu_5 + \sigma^4\mu_4 - \frac{\mu_4^2}{9} - \sigma^8 - \frac{4}{3}\sigma^2\mu_3^2. \tag{15}$$

Under symmetry, $\mu_3 = \mu_5 = 0$, so this reduces to

$$\tau_4^2 = \frac{\mu_8}{36} + \frac{2}{3}\sigma^2\mu_6 + \sigma^4\mu_4 - \frac{\mu_4^2}{9} - \sigma^8. \tag{16}$$

PROOF. Write

$$\psi_4(X) = A + B + C + D,$$

where

$$A = \frac{(X-\mu)^4}{12}, \qquad B = \frac{\sigma^2(X-\mu)^2}{2}, \qquad C = -\frac{\mu_3(X-\mu)}{3}, \qquad D = -\frac{\mu_4}{12} - \frac{\sigma^4}{2}.$$

Since $\mathbb{E}[\psi_4(X)] = 0$, we have $\mathrm{Var}(\psi_4(X)) = \mathbb{E}[\psi_4(X)^2]$. Expanding $(A+B+C+D)^2$, taking expectations term by term, and collecting like moments yields (15). This routine algebraic expansion can also be verified directly by symbolic computation software. The symmetric reduction follows immediately by setting $\mu_3 = \mu_5 = 0$. □

A.7. Bootstrap consistency for SPFD.

PROOF OF THE SUFFICIENCY PART OF THEOREM 11. Assume $\mathbb{E}|X-\mu|^{12} < \infty$. Since $\psi_4(X)$ has quartic tail growth, this implies

$$\mathbb{E}|\psi_4(X)|^3 < \infty.$$

The bootstrap Hoeffding decomposition gives

$$\sqrt{n}\,(\mathrm{SPFD}_n^* - \mathrm{SPFD}_n) = \frac{2}{\sqrt{n}} \sum_{i=1}^{n} \big(\psi_4(X_i^*) - \bar{\psi}_{4,n}\big) + o_{p^*}(1),$$

in probability, where

$$\bar{\psi}_{4,n} = \frac{1}{n} \sum_{i=1}^{n} \psi_4(X_i).$$

The finite third absolute moment yields conditional Lyapunov control for the self-normalized bootstrap projection sum. In addition, the bootstrap variance estimator $\widehat{\tau}_{4,n}^{*2}$ is conditionally consistent for τ_4^2. Therefore

$$T_{4,n}^* = \frac{\sqrt{n}(\mathrm{SPFD}_n^* - \mathrm{SPFD}_n)}{\widehat{\tau}_{4,n}^*} \xrightarrow{d} N(0,1)$$

conditionally in probability. □

A.8. Sharpness under regularly varying tails.

LEMMA 5. *Assume the distribution of $|X-\mu|$ has a regularly varying right tail with index $\alpha > 0$, that is,*

$$\Pr(|X-\mu| > x) = x^{-\alpha} L(x),$$

where L is slowly varying at infinity. Let

$$M_n = \max_{1 \le i \le n} |X_i - \mu|.$$

Then there exists a positive sequence $a_n \to \infty$ such that

$$a_n^{-1} M_n \xrightarrow{d} Z,$$

where Z is a non-degenerate Fréchet random variable.

PROOF. The convergence of $a_n^{-1}M_n$ to a Fréchet limit is a standard consequence of classical extreme-value theory for regularly varying tails; see, for example, Theorem 1.5.3 in Embrechts et al. [3] or the discussion in Resnick [10, Chapter 6]. Since $a_n \asymp n^{1/\alpha}\ell(n)$ for a slowly varying function ℓ, it follows that

$$M_n^{12}/n = O_p\big(n^{12/\alpha-1}\ell(n)^{12}\big),$$

and therefore $M_n^{12}/n \to \infty$ in probability whenever $\alpha < 12$. □

LEMMA 6. *Assume the distribution of $|X-\mu|$ has a regularly varying right tail with index $\alpha \in (8,12)$. Then*

$$\frac{1}{n}\sum_{i=1}^{n}|X_i-\mu|^{12} \xrightarrow{p} \infty.$$

PROOF. Since

$$M_n^{12} \le \sum_{i=1}^{n}|X_i-\mu|^{12},$$

it suffices to show that $n^{-1}M_n^{12} \to \infty$ in probability. By Lemma 5, we have $M_n = a_n O_p(1)$, where $a_n \asymp n^{1/\alpha}\ell(n)$ for a slowly varying function ℓ. Hence

$$\frac{M_n^{12}}{n} \asymp_p n^{12/\alpha-1}\ell(n)^{12} \to \infty,$$

because $\alpha < 12$. The conclusion follows. □

LEMMA 7. *Assume the distribution of $X-\mu$ has a regularly varying right tail with index $\alpha \in (8,12)$. Then*

$$\frac{\frac{1}{n}\sum_{i=1}^{n}|\psi_4(X_i)|^3}{\big(\frac{1}{n}\sum_{i=1}^{n}\psi_4(X_i)^2\big)^{3/2}} \xrightarrow{p} \infty.$$

Consequently, the conditional Lyapunov condition for the bootstrap empirical projection array fails with probability tending to one.

PROOF. Since $\psi_4(x)$ is a polynomial in $(x-\mu)$ with quartic leading term, we have

$$|\psi_4(x)|^3 \asymp |x-\mu|^{12} \qquad \text{as } |x| \to \infty.$$

By Lemma 6,

$$\frac{1}{n}\sum_{i=1}^{n}|\psi_4(X_i)|^3 \xrightarrow{p} \infty.$$

Also, since $\alpha > 8$, we have $\mathbb{E}[\psi_4(X)^2] < \infty$ by Remark 4. Hence

$$\frac{1}{n}\sum_{i=1}^{n}\psi_4(X_i)^2 \xrightarrow{p} \mathbb{E}[\psi_4(X)^2] \in (0,\infty)$$

by the law of large numbers. Therefore the displayed ratio diverges in probability, which is exactly the failure of the conditional Lyapunov condition for the bootstrap empirical projection array. □

PROOF OF THE NECESSITY PART OF THEOREM 11. Assume the distribution of $X-\mu$ has a regularly varying right tail with index $\alpha \in (8,12)$. Then $\mathbb{E}|X-\mu|^8 < \infty$ but $\mathbb{E}|X-\mu|^{12} = \infty$. Hence the central limit theorem for SPFD_n still holds, since $\mathbb{E}[\psi_4(X)^2] < \infty$. However, by Lemma 7, the Lyapunov ratio for the bootstrap empirical projection array diverges in probability. Thus the conditional Lyapunov condition fails with probability tending to one, so the bootstrap self-normalized projection term cannot satisfy a conditional central limit theorem. Because the studentized bootstrap statistic is asymptotically governed by this term, its conditional law cannot converge to the standard normal distribution. Therefore there exists $\varepsilon > 0$ such that

$$\limsup_{n\to\infty} \Pr\left(\sup_{t\in\mathbb{R}} \left| \overset{*}{\Pr}(T^*_{4,n} \le t) - \Phi(t) \right| > \varepsilon \right) > 0.$$

This proves invalidity of the studentized bootstrap throughout the regularly varying regime $\alpha \in (8,12)$. □

A.9. Bootstrap failure analysis. This subsection records the general mechanism behind the breakdown of the ordinary n-out-of-n bootstrap below the relevant projection-moment threshold. Let

$$U_n = \binom{n}{2}^{-1} \sum_{1\le i<j\le n} h(X_i, X_j)$$

be a non-degenerate second-order U-statistic with symmetric kernel h and parameter

$$\theta = \mathbb{E}[h(X_1, X_2)].$$

Let

$$\psi(x) = \mathbb{E}[h(x, X_2)] - \theta$$

denote the first-order Hájek projection. At the projection level, bootstrap validity requires that the empirical bootstrap projection array satisfy a conditional central limit theorem. A necessary ingredient is a conditional Lyapunov condition for the self-normalized projection array, so the decisive comparison is between the second and third absolute moments of $\psi(X)$.

Assume that there exist constants $c_1, c_2 > 0$ and $x_0 < \infty$ such that

$$c_1|x|^m \le |\psi(x)| \le c_2|x|^m \qquad \text{for all } |x| \ge x_0,$$

and suppose that X has a regularly varying tail with index $\alpha > 0$, that is,

$$\Pr(|X| > x) = x^{-\alpha} L(x),$$

where L is slowly varying at infinity. Then $\mathbb{E}[\psi(X)^2] < \infty$ whenever $\alpha > 2m$, whereas $\mathbb{E}|\psi(X)|^3 = \infty$ whenever $\alpha < 3m$. In the intermediate regime $2m < \alpha < 3m$, the empirical third absolute moment of the projection diverges. Indeed, if

$$M_n = \max_{1\le i\le n} |X_i|,$$

then regular variation implies that M_n is of order $n^{1/\alpha}$ up to slowly varying factors, and

$$\frac{1}{n}\sum_{i=1}^n |\psi(X_i)|^3 \ge \frac{c_1^3}{n} M_n^{3m}\, \mathbf{1}\{M_n \ge x_0\}.$$

Hence

$$\frac{1}{n}\sum_{i=1}^n |\psi(X_i)|^3 = \Omega_p\big(n^{3m/\alpha-1}\ell(n)^{3m}\big),$$

for a slowly varying function ℓ, and therefore

$$\frac{1}{n}\sum_{i=1}^{n}|\psi(X_i)|^3 \xrightarrow{p} \infty.$$

In particular, the empirical measure assigns mass $1/n$ to observations of order M_n, and these extreme points dominate the projection moments whenever $\alpha < 3m$.

Consequently, the empirical bootstrap projection array fails the conditional Lyapunov condition with probability tending to one. Therefore the empirical bootstrap projection array cannot satisfy a conditional central limit theorem in this regime. This is the structural obstruction isolated in Theorem 12: bootstrap failure is caused not merely by the loss of a nominal moment condition, but by domination of the empirical projection array by extreme order statistics.

For SPDV_n, the projection is quadratic ($m = 2$), yielding the fourth/eighth separation between asymptotic normality and studentized bootstrap validity. For SPFD_n, the projection is quartic ($m = 4$), and the same mechanism produces the sharp eighth/twelfth separation established in Theorem 11. More generally, for even pairwise kernels of degree $2k$ (so that $m = 2k$), this mechanism yields the $4k/6k$ separation predicted by the projection-growth principle. Modified procedures such as m-out-of-n bootstrap, subsampling, or tail-truncated kernels may restore validity by reducing the influence of extreme observations, but that lies outside the scope of the present paper.

A.10. Proof idea for Theorem 12. The argument follows the same structure as the necessity proof for Theorem 11. If the first-order projection satisfies $|\psi(x)| \asymp |x|^m$, then under regular variation with tail index α, the projection has finite second moment exactly when $\alpha > 2m$, and finite third absolute moment only when $\alpha > 3m$. In the intermediate regime $2m < \alpha < 3m$, the empirical third absolute moment

$$\frac{1}{n}\sum_{i=1}^{n}|\psi(X_i)|^3$$

is dominated by the sample maximum and diverges in probability. Thus the conditional Lyapunov ratio for the bootstrap projection array cannot remain tight. The bootstrap linear term is therefore governed by extremes rather than by a central limit effect, which is the mechanism behind the failure described in Theorem 12.

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